

```
In [663... import pandas as pd
qvi = pd.read_csv('qvi_data.csv')
```

```
In [664... qvi = qvi[~qvi['PROD_NAME'].str.contains('SALSA',case=False)]
```

```
In [665... qvi.drop_duplicates(inplace=True)
```

```
In [666... qvi['DATE'] = pd.to_datetime(qvi['DATE'])

qvi['Month_id']=qvi['DATE'].dt.to_period('M')
grouped = qvi.groupby(['STORE_NBR'])['Month_id'].nunique()
not_all_tns = grouped[grouped!=12].index
all_tns = grouped[grouped==12].index

qvi.drop(qvi[qvi['STORE_NBR'].isin(not_all_tns)].index, inplace=True)

print(not_all_tns)
print(all_tns)
```

```
Index([11, 31, 44, 76, 85, 92, 117, 177, 193, 206, 218, 252], dtype='int64', name='STORE_NBR')
```

```
Index([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10,
      ...
      263, 264, 265, 266, 267, 268, 269, 270, 271, 272],
      dtype='int64', name='STORE_NBR', length=259)
```

```
In [667... qvi.head()
```

Out[667...

	LYLTY_CARD_NBR	DATE	STORE_NBR	TXN_ID	PROD_NBR	PROD_NAME	PROD_QTY	TOT_SALES	PACK_SIZE	BRAND
0	1000	2018-10-17	1	1	5	Natural Chip Compny SeaSalt175g	2	6.0	175	NATURAL
1	1002	2018-09-16	1	2	58	Red Rock Deli Chikn&Garlic Aioli 150g	1	2.7	150	RRR
2	1003	2019-03-07	1	3	52	Grain Waves Sour Cream&Chives 210G	1	3.6	210	GRNWVE
3	1003	2019-03-08	1	4	106	Natural ChipCo Hony Soy Chckn175g	1	3.0	175	NATURAL
4	1004	2018-11-02	1	5	96	WW Original Stacked Chips 160g	1	1.9	160	WOOLWORTH



In [668... *# for analysis we store the matrices in single data frame*
easy for correlation and plotting

```
In [669... Measure_over_time = qvi.groupby(['STORE_NBR', 'Month_id']).agg({'TOT_SALES': 'sum',
                                                                'LYLTY_CARD_NBR': pd.Series.nunique,
                                                                'TXN_ID': 'count',
                                                                'PROD_QTY': 'sum'}).reset_index()

Measure_over_time = Measure_over_time.rename(columns={'LYLTY_CARD_NBR': 'NUM_CUS',
                                                    'TXN_ID': 'NUM_TNS'})

Measure_over_time
```

Out[669...

	STORE_NBR	Month_id	TOT_SALES	NUM_CUS	NUM_TNS	PROD_QTY
0	1	2018-07	188.9	47	49	58
1	1	2018-08	168.4	41	41	52
2	1	2018-09	268.1	57	59	71
3	1	2018-10	175.4	39	40	51
4	1	2018-11	184.8	44	45	55
...	...	...	...	...	...	...
3103	272	2019-02	385.3	44	47	89
3104	272	2019-03	421.9	48	51	97
3105	272	2019-04	445.1	54	56	105
3106	272	2019-05	314.6	34	40	71
3107	272	2019-06	301.9	33	36	68

3108 rows × 6 columns

In [670...

type(Measure_over_time)

Out[670...

pandas.core.frame.DataFrame

In [671...

Measure_over_time ['TXN_PER_CUS'] = (Measure_over_time['NUM_TNS'] / Measure_over_time['NUM_CUS'])

Measure_over_time

Out[671...

	STORE_NBR	Month_id	TOT_SALES	NUM_CUS	NUM_TNS	PROD_QTY	TXN_PER_CUS
0	1	2018-07	188.9	47	49	58	1.042553
1	1	2018-08	168.4	41	41	52	1.000000
2	1	2018-09	268.1	57	59	71	1.035088
3	1	2018-10	175.4	39	40	51	1.025641
4	1	2018-11	184.8	44	45	55	1.022727
...	...	...	...	...	...	...	...
3103	272	2019-02	385.3	44	47	89	1.068182
3104	272	2019-03	421.9	48	51	97	1.062500
3105	272	2019-04	445.1	54	56	105	1.037037
3106	272	2019-05	314.6	34	40	71	1.176471
3107	272	2019-06	301.9	33	36	68	1.090909

3108 rows × 7 columns

In [672...

```
Measure_over_time ['AVG_PRICE'] = (Measure_over_time['TOT_SALES'] / Measure_over_time['PROD_QTY'])
Measure_over_time
```

Out[672...

	STORE_NBR	Month_id	TOT_SALES	NUM_CUS	NUM_TNS	PROD_QTY	TXN_PER_CUS	AVG_PRICE
0	1	2018-07	188.9	47	49	58	1.042553	3.256897
1	1	2018-08	168.4	41	41	52	1.000000	3.238462
2	1	2018-09	268.1	57	59	71	1.035088	3.776056
3	1	2018-10	175.4	39	40	51	1.025641	3.439216
4	1	2018-11	184.8	44	45	55	1.022727	3.360000
...	...	...	...	...	...	...	...	...
3103	272	2019-02	385.3	44	47	89	1.068182	4.329213
3104	272	2019-03	421.9	48	51	97	1.062500	4.349485
3105	272	2019-04	445.1	54	56	105	1.037037	4.239048
3106	272	2019-05	314.6	34	40	71	1.176471	4.430986
3107	272	2019-06	301.9	33	36	68	1.090909	4.439706

3108 rows × 8 columns

In [673...

```
# it have measures before implementing trial store
preTrailMeasures = Measure_over_time[(Measure_over_time['Month_id'] < pd.Period('2019-02', freq='M'))]
preTrailMeasures
```

Out[673...

	STORE_NBR	Month_id	TOT_SALES	NUM_CUS	NUM_TNS	PROD_QTY	TXN_PER_CUS	AVG_PRICE
0	1	2018-07	188.9	47	49	58	1.042553	3.256897
1	1	2018-08	168.4	41	41	52	1.000000	3.238462
2	1	2018-09	268.1	57	59	71	1.035088	3.776056
3	1	2018-10	175.4	39	40	51	1.025641	3.439216
4	1	2018-11	184.8	44	45	55	1.022727	3.360000
...	...	...	...	...	...	...	...	...
3098	272	2018-09	294.5	31	35	69	1.129032	4.268116
3099	272	2018-10	405.1	41	48	94	1.170732	4.309574
3100	272	2018-11	355.8	39	43	83	1.102564	4.286747
3101	272	2018-12	363.1	43	43	81	1.000000	4.482716
3102	272	2019-01	392.4	44	47	90	1.068182	4.360000

1813 rows × 8 columns

In [674...

```

# we create a correlation func with 3 input parameters [inputTable, metricCol, storecomparision]
# inputTable - contain store metrics
# metricCol - Which metric column we compare
# storecomparision - trial store number

# Correlation Score → Checks whether the trend/pattern is similar.
# Magnitude Score → Checks whether the actual values (sales, customers) are similar.

# we create table with store-1, store-2, corr_measure, magnitude

```

In [675...

```

def find (input_table, metric_col,store_comparision) :

    pivot_table = input_table.pivot(index = 'Month_id',
                                      columns = 'STORE_NBR',
                                      values = metric_col)

```

```
metric_table = pivot_table.corr()[store_comparision].reset_index()
metric_table.columns = ['STORE-2', 'CORR_MEASURE']
metric_table['STORE-1'] = store_comparision
metric_table = metric_table[['STORE-1', 'STORE-2', 'CORR_MEASURE']]

trail_store = pivot_table[store_comparision]
result=[]
for i in pivot_table.columns :
    current_store = pivot_table[i]

    diff = abs(trail_store-current_store)

    avg_diff = diff.mean()

    result.append({'AVG_DIFF':avg_diff})

df=pd.DataFrame(result)

metric_table = metric_table.join(df)

min_diff=metric_table['AVG_DIFF'].min()
max_diff=metric_table['AVG_DIFF'].max()

metric_table['MAG_SCORE'] = 1-((metric_table['AVG_DIFF']-min_diff)/(max_diff-min_diff))
metric_table['COMPOSITE_VAL'] = ((metric_table['CORR_MEASURE']+metric_table['MAG_SCORE'])/2)

metric_table = metric_table[['STORE-1', 'STORE-2', 'COMPOSITE_VAL']]

return metric_table

compo_sales_77 = find(preTrailMeasures, 'TOT_SALES',77)
compo_sales_77
```

Out[675...

	STORE-1	STORE-2	COMPOSITE_VAL
0	77	1	0.472365
1	77	2	0.340367
2	77	3	0.483611
3	77	4	-0.106848
4	77	5	0.199870
...	...	...	...
254	77	268	0.677968
255	77	269	-0.022732
256	77	270	0.350941
257	77	271	0.369656
258	77	272	0.352212

259 rows × 3 columns

In [676...

calling the function to find corr for monthly no. of customer

```
compo_cus_77 = find(preTrailMeasures, 'NUM_CUS',77)  
compo_cus_77
```

Out[676...

	STORE-1	STORE-2	COMPOSITE_VAL
0	77	1	0.637337
1	77	2	0.154361
2	77	3	0.533233
3	77	4	-0.070557
4	77	5	0.357250
...	...	...	...
254	77	268	0.654852
255	77	269	0.041297
256	77	270	0.177084
257	77	271	0.260633
258	77	272	0.507482

259 rows × 3 columns

```
In [677... #for the trial store 86,88 (call func to find composite value which is addition of correlation and magnitude) both sc
compo_sales_86 = find(preTrailMeasures, 'TOT_SALES',86)
compo_cus_86 = find(preTrailMeasures, 'NUM_CUS',86)

compo_sales_88 = find(preTrailMeasures, 'TOT_SALES',88)
compo_cus_88 = find(preTrailMeasures, 'NUM_CUS',88)
```

```
In [678... # we merge the sales and customer calculation for the store trial store

final_df_77 = pd.merge(compo_cus_77,compo_cus_77,on=['STORE-1','STORE-2'])
final_df_77=final_df_77.rename(columns={'COMPOSITE_VAL_x':'SALES_SCORE',
                                         'COMPOSITE_VAL_y':'CUSTOMER_SCORE'})

final_df_86 = pd.merge(compo_cus_86,compo_cus_86,on=['STORE-1','STORE-2'])
final_df_86=final_df_86.rename(columns={'COMPOSITE_VAL_x':'SALES_SCORE',
                                         'COMPOSITE_VAL_y':'CUSTOMER_SCORE'})
```

```
final_df_88 = pd.merge(compo_cus_88, compo_cus_88, on=['STORE-1', 'STORE-2'])
final_df_88 = final_df_88.rename(columns={'COMPOSITE_VAL_x': 'SALES_SCORE',
                                         'COMPOSITE_VAL_y': 'CUSTOMER_SCORE'})

# final control score is calculated by ADDING the sales and customer COMPOSITE value

final_df_77['FinalControlScore_77'] = (final_df_77['SALES_SCORE'] * 0.5) + (final_df_77['CUSTOMER_SCORE'] * 0.5)
print(final_df_77)

final_df_86['FinalControlScore_86'] = (final_df_86['SALES_SCORE'] * 0.5) + (final_df_86['CUSTOMER_SCORE'] * 0.5)
print(final_df_86)

final_df_88['FinalControlScore_88'] = (final_df_88['SALES_SCORE'] * 0.5) + (final_df_88['CUSTOMER_SCORE'] * 0.5)
final_df_88
```

	STORE-1	STORE-2	SALES_SCORE	CUSTOMER_SCORE	FinalControlScore_77
0	77	1	0.637337	0.637337	0.637337
1	77	2	0.154361	0.154361	0.154361
2	77	3	0.533233	0.533233	0.533233
3	77	4	-0.070557	-0.070557	-0.070557
4	77	5	0.357250	0.357250	0.357250
..	...	...	...	...	...
254	77	268	0.654852	0.654852	0.654852
255	77	269	0.041297	0.041297	0.041297
256	77	270	0.177084	0.177084	0.177084
257	77	271	0.260633	0.260633	0.260633
258	77	272	0.507482	0.507482	0.507482

[259 rows x 5 columns]

	STORE-1	STORE-2	SALES_SCORE	CUSTOMER_SCORE	FinalControlScore_86
0	86	1	0.406918	0.406918	0.406918
1	86	2	0.142227	0.142227	0.142227
2	86	3	0.490030	0.490030	0.490030
3	86	4	0.386026	0.386026	0.386026
4	86	5	0.500891	0.500891	0.500891
..	...	...	...	...	...
254	86	268	0.188343	0.188343	0.188343
255	86	269	0.618644	0.618644	0.618644
256	86	270	0.084341	0.084341	0.084341
257	86	271	0.590724	0.590724	0.590724
258	86	272	-0.012653	-0.012653	-0.012653

[259 rows x 5 columns]

Out[678...

	STORE-1	STORE-2	SALES_SCORE	CUSTOMER_SCORE	FinalControlScore_88
0	88	1	0.382867	0.382867	0.382867
1	88	2	-0.137336	-0.137336	-0.137336
2	88	3	0.638252	0.638252	0.638252
3	88	4	0.357066	0.357066	0.357066
4	88	5	0.339460	0.339460	0.339460
...	...	...	...	...	...
254	88	268	0.426286	0.426286	0.426286
255	88	269	0.381082	0.381082	0.381082
256	88	270	0.363865	0.363865	0.363865
257	88	271	0.294082	0.294082	0.294082
258	88	272	0.091676	0.091676	0.091676

259 rows × 5 columns

In [679...

```

# to find the best control store we dicard the trial store and list the val in descending order and take the top val
final_df_77=final_df_77[final_df_77['STORE-2']!=77]
final_df_77=final_df_77.sort_values(by='FinalControlScore_77',ascending=False)
print(final_df_77)

final_df_86=final_df_86[final_df_86['STORE-2']!=86]
final_df_86=final_df_86.sort_values(by='FinalControlScore_86',ascending=False)
print(final_df_86)

final_df_88=final_df_88[final_df_88['STORE-2']!=88]
final_df_88=final_df_88.sort_values(by='FinalControlScore_88',ascending=False)
final_df_88

```

	STORE-1	STORE-2	SALES_SCORE	CUSTOMER_SCORE	FinalControlScore_77
220	77	233	0.978102	0.978102	0.978102
240	77	254	0.913680	0.913680	0.913680
32	77	35	0.894238	0.894238	0.894238
79	77	84	0.884686	0.884686	0.884686
25	77	27	0.857479	0.857479	0.857479
..	...	...	...	...	...
157	77	165	-0.112863	-0.112863	-0.112863
139	77	147	-0.116532	-0.116532	-0.116532
95	77	102	-0.117923	-0.117923	-0.117923
71	77	75	-0.128480	-0.128480	-0.128480
214	77	227	-0.174683	-0.174683	-0.174683

[258 rows x 5 columns]

	STORE-1	STORE-2	SALES_SCORE	CUSTOMER_SCORE	FinalControlScore_86
107	86	114	0.905042	0.905042	0.905042
147	86	155	0.900589	0.900589	0.900589
212	86	225	0.875970	0.875970	0.875970
73	86	78	0.827978	0.827978	0.827978
130	86	138	0.776861	0.776861	0.776861
..	...	...	...	...	...
153	86	161	-0.190928	-0.190928	-0.190928
39	86	42	-0.235987	-0.235987	-0.235987
119	86	127	-0.236332	-0.236332	-0.236332
138	86	146	-0.253800	-0.253800	-0.253800
183	86	192	-0.311992	-0.311992	-0.311992

[258 rows x 5 columns]

Out[679...

	STORE-1	STORE-2	SALES_SCORE	CUSTOMER_SCORE	FinalControlScore_88
224	88	237	0.968539	0.968539	0.968539
169	88	178	0.857819	0.857819	0.857819
53	88	57	0.783717	0.783717	0.783717
115	88	123	0.771275	0.771275	0.771275
149	88	157	0.765240	0.765240	0.765240
...	...	...	...	...	...
253	88	267	-0.179374	-0.179374	-0.179374
177	88	186	-0.179966	-0.179966	-0.179966
226	88	239	-0.209768	-0.209768	-0.209768
252	88	266	-0.286146	-0.286146	-0.286146
244	88	258	-0.364666	-0.364666	-0.364666

258 rows × 5 columns

In [680...

```
# best CONTROL STORE for the trial stores
print(final_df_77['STORE-2'].head(1))

print('\n',final_df_86['STORE-2'].head(1))

final_df_88['STORE-2'].head(1)
```

```
220    233
Name: STORE-2, dtype: int64
```

```
107    114
Name: STORE-2, dtype: int64
```

Out[680...

```
224    237
Name: STORE-2, dtype: int64
```

In [681...

```
type(Measure_over_time)
```

Out[681... pandas.core.frame.DataFrame

```
In [682... # we create the store_type col for the trial store 77
Measure_over_time['STORE_TYPE'] = 'other'
Measure_over_time.loc[Measure_over_time['STORE_NBR']==77,'STORE_TYPE'] = 'Trial'

Measure_over_time.loc[Measure_over_time['STORE_NBR']==233,'STORE_TYPE'] = 'Control'

Measure_over_time = Measure_over_time[['STORE_NBR','Month_id','TOT_SALES','STORE_TYPE','NUM_CUS']]

Measure_over_time
```

Out[682...

	STORE_NBR	Month_id	TOT_SALES	STORE_TYPE	NUM_CUS
0	1	2018-07	188.9	other	47
1	1	2018-08	168.4	other	41
2	1	2018-09	268.1	other	57
3	1	2018-10	175.4	other	39
4	1	2018-11	184.8	other	44
...	...	...	...	...	...
3103	272	2019-02	385.3	other	44
3104	272	2019-03	421.9	other	48
3105	272	2019-04	445.1	other	54
3106	272	2019-05	314.6	other	34
3107	272	2019-06	301.9	other	33

3108 rows × 5 columns

```
In [683... Measure_over_time = Measure_over_time.rename(columns={'STORE_TYPE':'STORE_TYPE_77'})
```

```
In [684... # created the store type col for the trial store 86 and 88
Measure_over_time['STORE_TYPE_86'] = 'other'
```

```

Measure_over_time['STORE_TYPE_88'] = 'other'

Measure_over_time.loc[Measure_over_time['STORE_NBR']==86, 'STORE_TYPE_86'] = 'Trial'
Measure_over_time.loc[Measure_over_time['STORE_NBR']==155, 'STORE_TYPE_86'] = 'Control'

Measure_over_time.loc[Measure_over_time['STORE_NBR']==88, 'STORE_TYPE_88'] = 'Trial'
Measure_over_time.loc[Measure_over_time['STORE_NBR']==237, 'STORE_TYPE_88'] = 'Control'

Measure_over_time = Measure_over_time[['STORE_NBR', 'Month_id', 'TOT_SALES', 'STORE_TYPE_77', 'STORE_TYPE_86', 'STORE_TYPE_88', 'NUM_CUS']]

Measure_over_time

```

Out[684]...

	STORE_NBR	Month_id	TOT_SALES	STORE_TYPE_77	STORE_TYPE_86	STORE_TYPE_88	NUM_CUS
0	1	2018-07	188.9	other	other	other	47
1	1	2018-08	168.4	other	other	other	41
2	1	2018-09	268.1	other	other	other	57
3	1	2018-10	175.4	other	other	other	39
4	1	2018-11	184.8	other	other	other	44
...	...	...	...	...	...	...	...
3103	272	2019-02	385.3	other	other	other	44
3104	272	2019-03	421.9	other	other	other	48
3105	272	2019-04	445.1	other	other	other	54
3106	272	2019-05	314.6	other	other	other	34
3107	272	2019-06	301.9	other	other	other	33

3108 rows × 7 columns

In [685]...

Measure_over_time

Out[685...

	STORE_NBR	Month_id	TOT_SALES	STORE_TYPE_77	STORE_TYPE_86	STORE_TYPE_88	NUM_CUS
0	1	2018-07	188.9	other	other	other	47
1	1	2018-08	168.4	other	other	other	41
2	1	2018-09	268.1	other	other	other	57
3	1	2018-10	175.4	other	other	other	39
4	1	2018-11	184.8	other	other	other	44
...	...	...	...	...	...	...	...
3103	272	2019-02	385.3	other	other	other	44
3104	272	2019-03	421.9	other	other	other	48
3105	272	2019-04	445.1	other	other	other	54
3106	272	2019-05	314.6	other	other	other	34
3107	272	2019-06	301.9	other	other	other	33

3108 rows × 7 columns

In [686...

```
# group the store type and dates then finding mean for total sales
```

```
Measure_over_time_77=Measure_over_time.groupby(['Month_id','STORE_TYPE_77']) ['TOT_SALES'].mean().reset_index()
```

```
Measure_over_time_86=Measure_over_time.groupby(['Month_id','STORE_TYPE_86']) ['TOT_SALES'].mean().reset_index()
```

```
Measure_over_time_88=Measure_over_time.groupby(['Month_id','STORE_TYPE_88']) ['TOT_SALES'].mean().reset_index()
```

In [687...

```
import matplotlib.pyplot as plt
def plotting (table,ST):
    plt.figure(figsize=(15,4))
    df=table[table[ST].isin(['Trial','Control'])]
    for storeT in df[ST].unique():
```

```

temp=df[df[ST]==storeT]

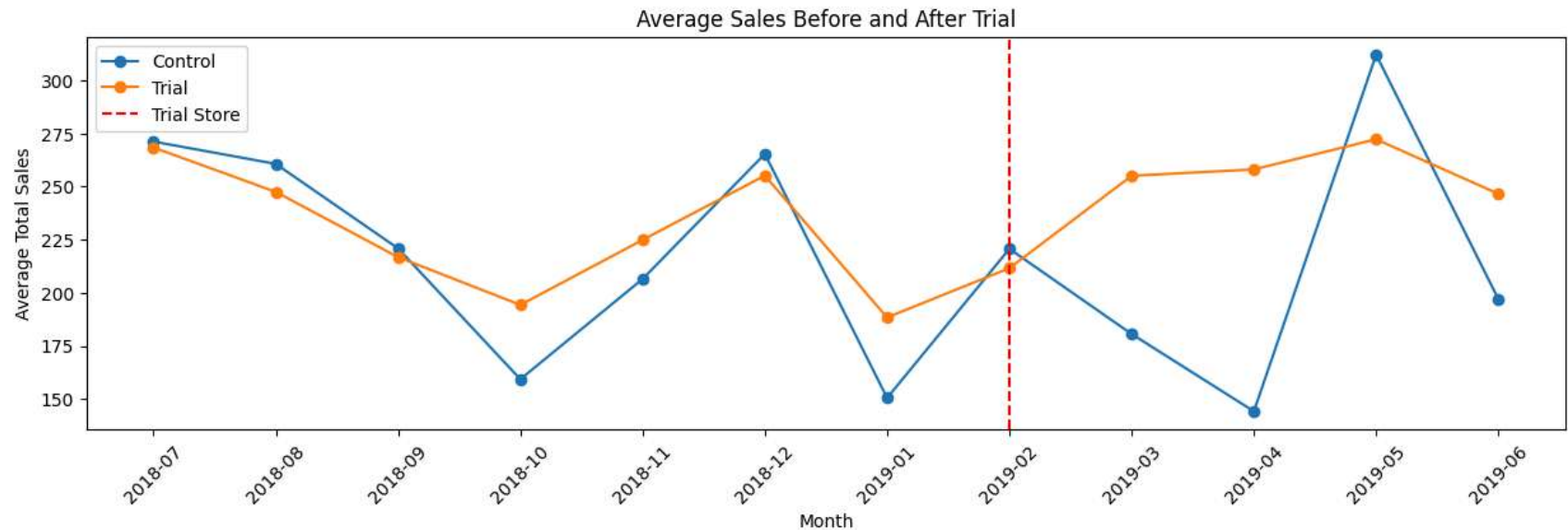
plt.plot(temp['Month_id'].astype(str),temp['TOT_SALES'],marker='o',label=storeT)

plt.axvline(x='2019-02',linestyle='--',label='Trial Store',color='red')

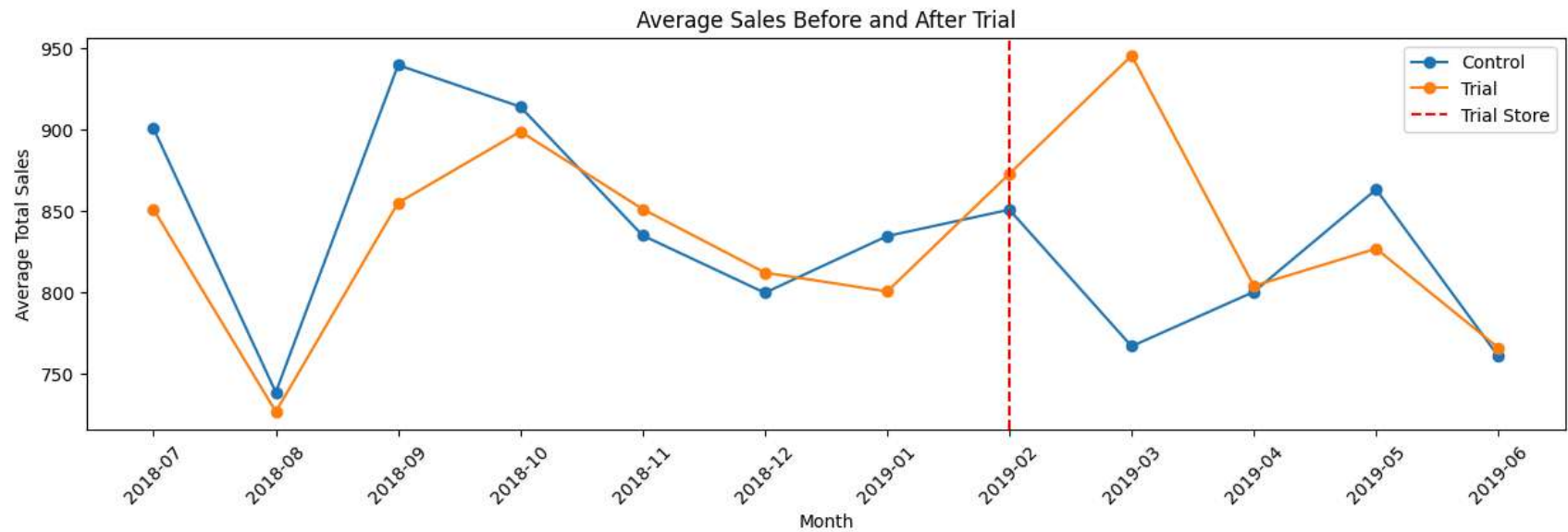
plt.title("Average Sales Before and After Trial")
plt.xlabel("Month")
plt.ylabel("Average Total Sales")
plt.legend()
plt.xticks(rotation=45)
plt.show()

```

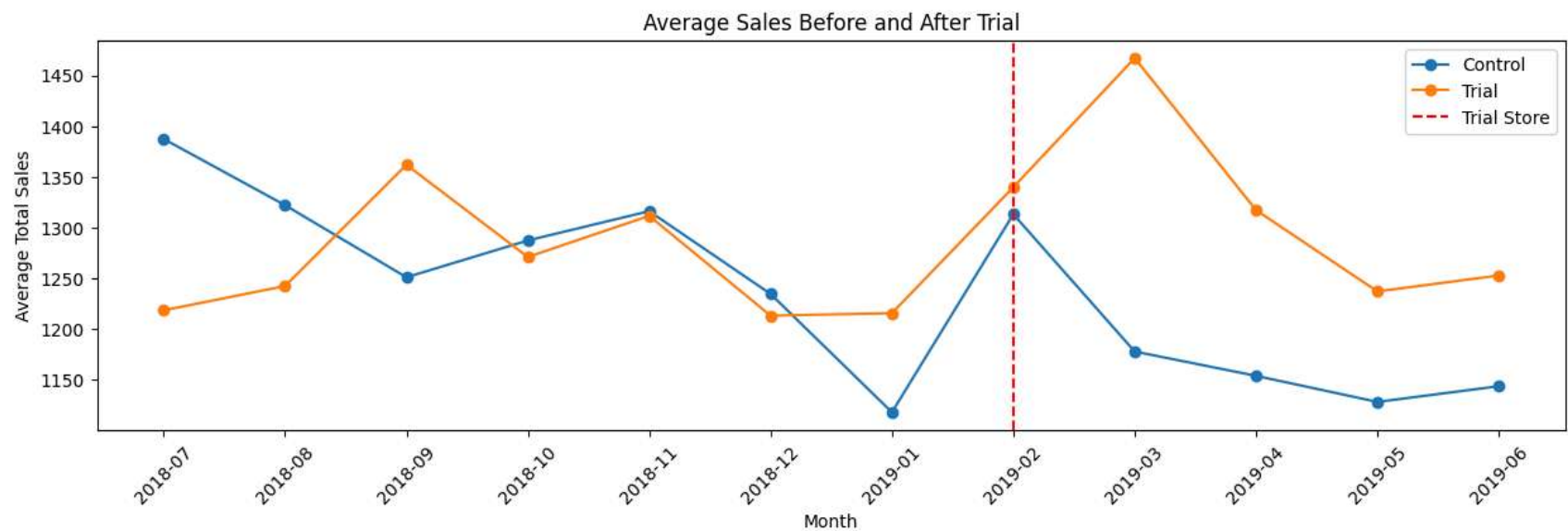
In [688... plotting(Measure_over_time_77,'STORE_TYPE_77')



In [689... plotting(Measure_over_time_86,'STORE_TYPE_86')



```
In [690... plotting(Measure_over_time_88, 'STORE_TYPE_88')
```



```
In [691... # pre trail months - 1 (before implementing trial store we have 6 months of data)
Dof = 7 - 1
```

```
In [692... from scipy.stats import t
critical_t = t.ppf(0.95,6)
critical_t
```

```
Out[692... np.float64(1.9431802805153018)
```

```
In [693... def insight(t,c,measure):
    # scale the control store to the trial store baseline

    scalefactor = preTrailMeasures.loc[(preTrailMeasures['STORE_NBR']==t) & (preTrailMeasures['Month_id'] < pd.Period(
    measurescaledsales=Measure_over_time.copy()
    trial = measurescaledsales.loc[measurescaledsales['STORE_NBR']==t,['STORE_NBR','Month_id', measure]].copy()
    control = measurescaledsales.loc[measurescaledsales['STORE_NBR']==c,['STORE_NBR','Month_id', measure]].copy()

    control['controlsales'] = control[measure]*scalefactor

    PD = trial.merge(control[['controlsales','Month_id']], on='Month_id',how = 'inner')

    PD['perc_diff'] = abs((PD[measure] - PD['controlsales']) / PD['controlsales']) * 100

    SD= PD.loc[PD['Month_id'] < pd.Period('2019-02',freq='M'),'perc_diff'].std()

    ''' Test Statistic=  $x - \mu$  / Standard Deviation
        x = Observed value (percentage difference)
         $\mu$  (mu) = Expected value under the null hypothesis (Zero)
        SD = Standard deviation '''

    PD['Ttest'] = PD['perc_diff'] / SD
    relative_sd = SD / 100

    Trial_assessment = PD.loc[(PD['Month_id'] >= pd.Period('2019-02',freq='M')) & (PD['Month_id'] < pd.Period('2019-02',freq='M'))

    # confidence level range is where the normal distribution is expected to fall 95% of the time.
    Trial_assessment['lower_confidence'] = (Trial_assessment['controlsales'] * (1 - 2 * relative_sd))
    Trial_assessment['upper_confidence'] = (Trial_assessment['controlsales'] * (1 + 2 * relative_sd))

    Trial_assessment = Trial_assessment[['Month_id', measure, 'controlsales', 'lower_confidence', 'upper_confidence']]

    plt.figure(figsize=(15,5))
    plt.plot(Trial_assessment['Month_id'].astype(str),Trial_assessment[measure],marker='o',color='#18966f',label = 'trial')
```

```

plt.plot(Trial_assessment['Month_id'].astype(str), Trial_assessment['controlsales'], marker='o', color='#20baba', label='Control Sales')
plt.plot(Trial_assessment['Month_id'].astype(str), Trial_assessment['lower_confidence'], linestyle='--', color='gray', label='Lower Confidence')
plt.plot(Trial_assessment['Month_id'].astype(str), Trial_assessment['upper_confidence'], linestyle='--', color='gray', label='Upper Confidence')

plt.title('IS the Trial Store Sales Significantly Different from Control Store?')
plt.xlabel('Month')
plt.ylabel('Sales')
plt.legend()

return PD, SD

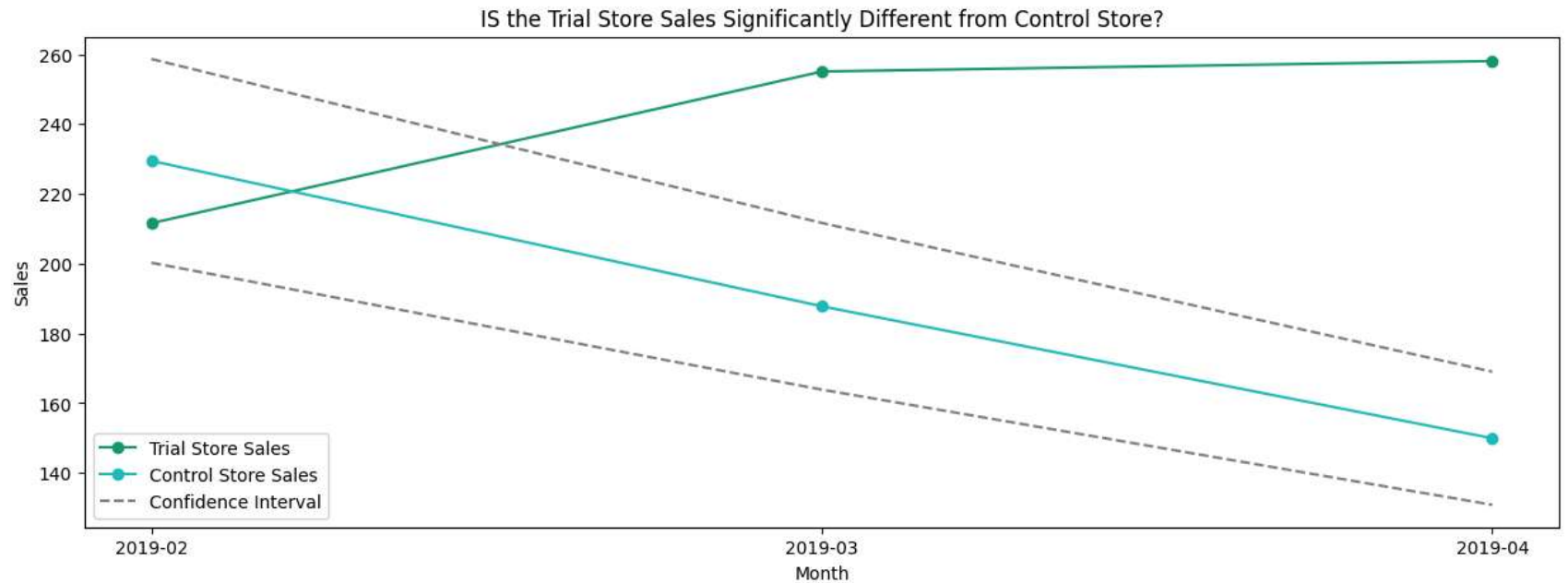
```

In [694... insight(77,233,'TOT_SALES')

```

Out[694... (
  STORE_NBR Month_id TOT_SALES controlsales perc_diff Ttest
0          77 2018-07      268.4      281.980841  4.816228  0.756805
1          77 2018-08      247.5      271.063441  8.692962  1.365980
2          77 2018-09      216.8      229.681297  5.608335  0.881273
3          77 2018-10      194.3      165.632551 17.307859  2.719694
4          77 2018-11      224.9      214.708863  4.746491  0.745846
5          77 2018-12      255.2      275.950277  7.519571  1.181598
6          77 2019-01      188.4      156.482731 20.396672  3.205059
7          77 2019-02      211.6      229.473346  7.788855  1.223912
8          77 2019-03      255.1      187.779277 35.850987  5.633494
9          77 2019-04      258.1      149.932291 72.144372 11.336505
10         77 2019-05      272.3      324.506712 16.088022  2.528014
11         77 2019-06      246.6      204.831215 20.391806  3.204295,
np.float64(6.363898833418711))

```

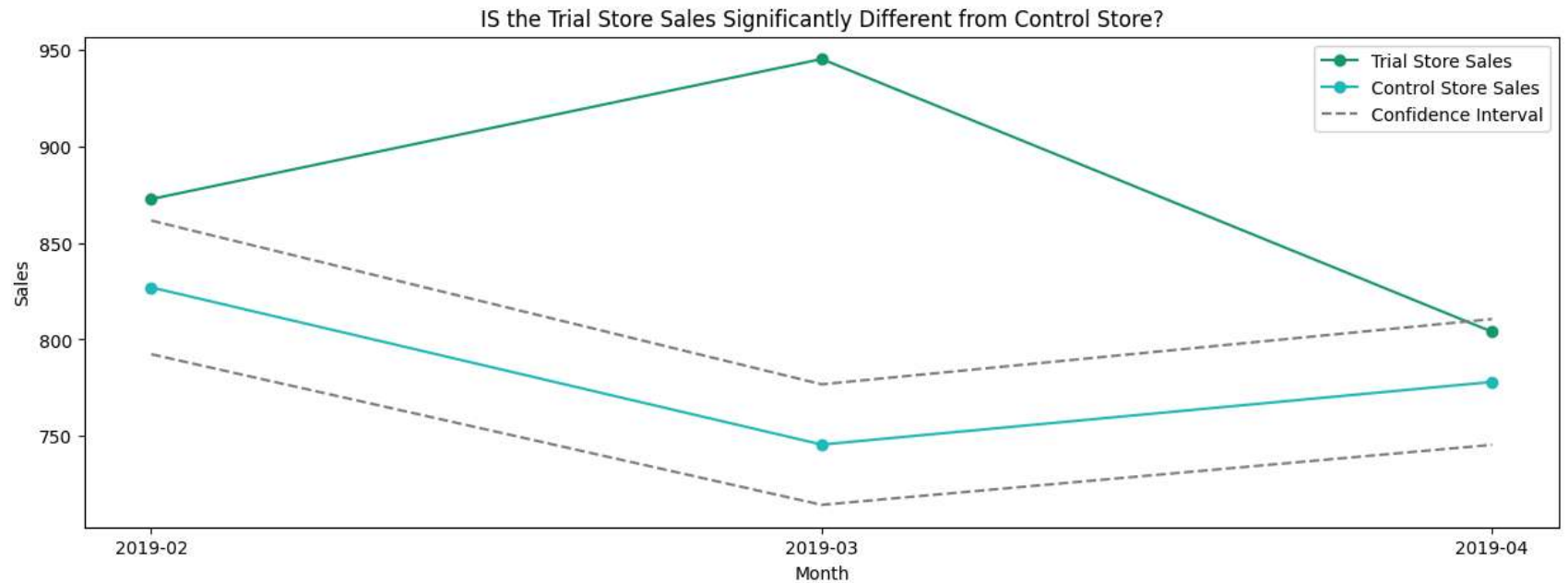


In [695... insight(86,155,'TOT_SALES')

Out[695... (

	STORE_NBR	Month_id	TOT_SALES	controlsales	perc_diff	Ttest
0	86	2018-07	851.00	875.427669	2.790370	1.332180
1	86	2018-08	726.85	718.052875	1.225136	0.584905
2	86	2018-09	855.00	913.337595	6.387298	3.049428
3	86	2018-10	898.80	888.453131	1.164594	0.556001
4	86	2018-11	851.20	811.661230	4.871339	2.325678
5	86	2018-12	812.20	777.445092	4.470400	2.134262
6	86	2019-01	800.60	811.272410	1.315515	0.628054
7	86	2019-02	872.80	827.019610	5.535587	2.642804
8	86	2019-03	945.40	745.561872	26.803695	12.796638
9	86	2019-04	804.00	778.028321	3.338141	1.593697
10	86	2019-05	826.90	839.121625	1.456478	0.695353
11	86	2019-06	766.00	739.535166	3.578577	1.708487,

np.float64(2.0945888446244103))

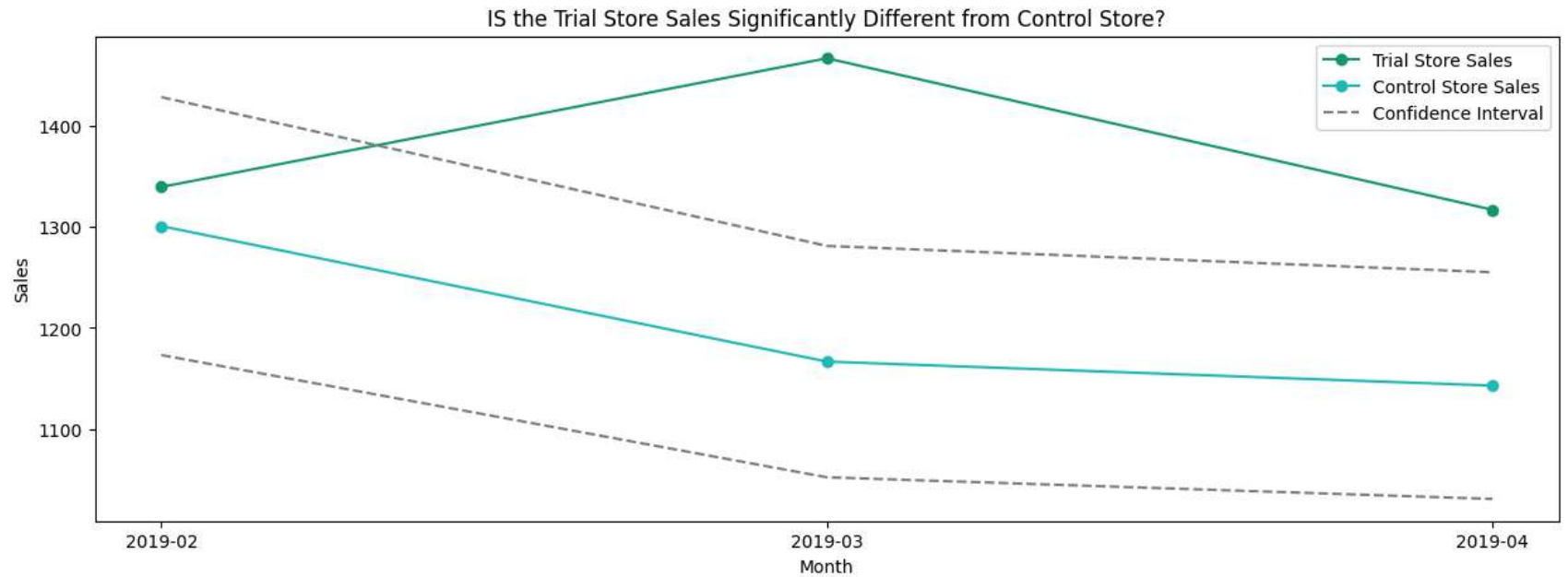


In [696... insight(88,237,'TOT_SALES')

Out[696... (

	STORE_NBR	Month_id	TOT_SALES	controlsales	perc_diff	Ttest
0	88	2018-07	1218.20	1374.394024	11.364574	2.315607
1	88	2018-08	1242.20	1309.696842	5.153623	1.050085
2	88	2018-09	1361.80	1239.253204	9.888762	2.014901
3	88	2018-10	1270.80	1275.218100	0.346458	0.070593
4	88	2018-11	1311.40	1303.851308	0.578953	0.117966
5	88	2018-12	1213.00	1223.004601	0.818035	0.166680
6	88	2019-01	1215.40	1107.381921	9.754365	1.987517
7	88	2019-02	1339.60	1300.879003	2.976526	0.606487
8	88	2019-03	1467.00	1166.728952	25.736144	5.243910
9	88	2019-04	1317.00	1142.950509	15.228086	3.102824
10	88	2019-05	1236.85	1117.487759	10.681302	2.176386
11	88	2019-06	1252.60	1132.844670	10.571205	2.153953,

np.float64(4.907815778187753))

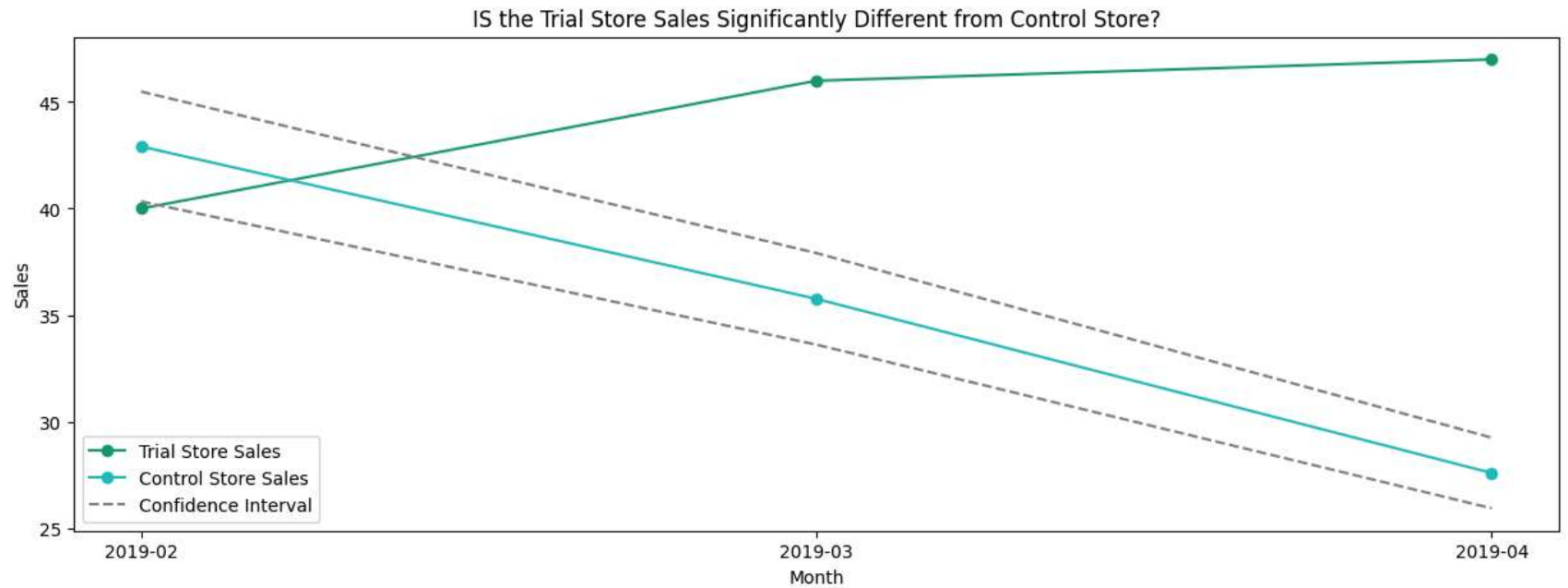


In [697... insight(77,233,'NUM_CUS')

Out[697... (

	STORE_NBR	Month_id	NUM_CUS	controlsales	perc_diff	Ttest
0	77	2018-07	47	48.021739	2.127660	0.708192
1	77	2018-08	46	44.956522	2.321083	0.772573
2	77	2018-09	40	40.869565	2.127660	0.708192
3	77	2018-10	36	32.695652	10.106383	3.363912
4	77	2018-11	39	39.847826	2.127660	0.708192
5	77	2018-12	43	43.934783	2.127660	0.708192
6	77	2019-01	31	31.673913	2.127660	0.708192
7	77	2019-02	40	42.913043	6.788247	2.259470
8	77	2019-03	46	35.760870	28.632219	9.530240
9	77	2019-04	47	27.586957	70.370370	23.422793
10	77	2019-05	53	55.173913	3.940110	1.311467
11	77	2019-06	38	34.739130	9.386733	3.124376,

np.float64(3.00435434917479))

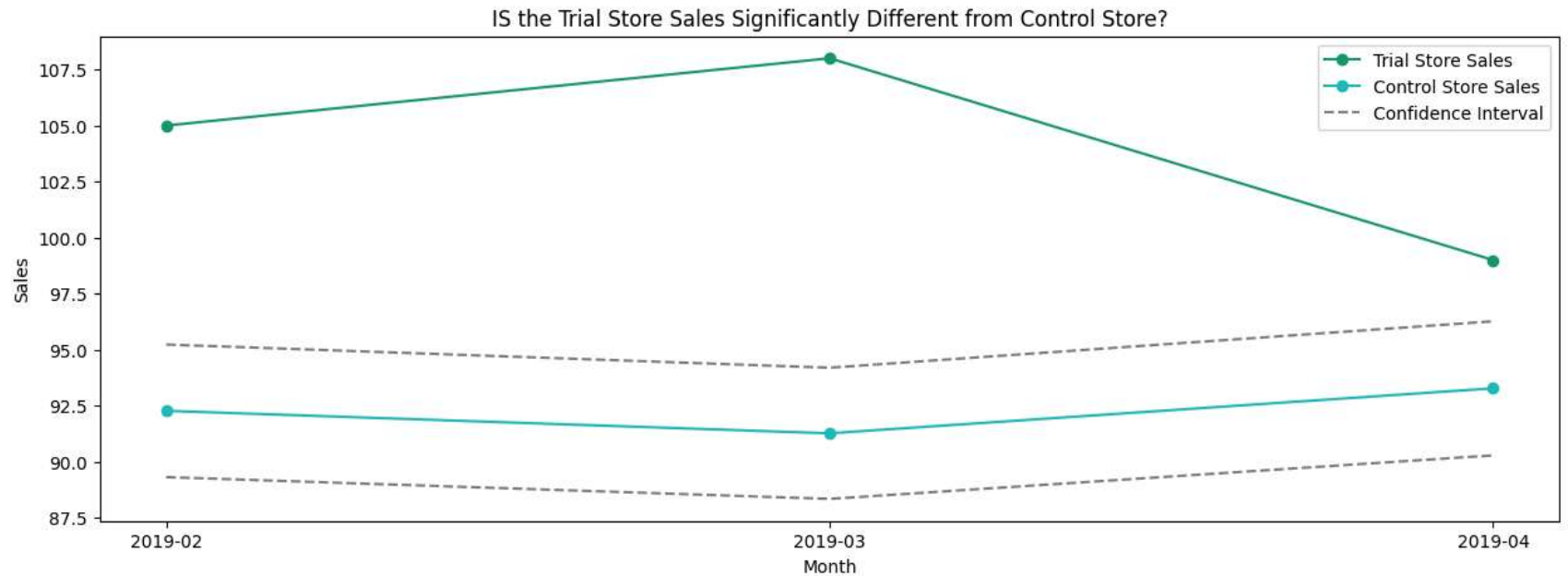


In [698... `insight(86,155,'NUM_CUS')`

Out[698... (

	STORE_NBR	Month_id	NUM_CUS	controlsales	perc_diff	Ttest
0	86	2018-07	94	98.294294	4.368813	2.726632
1	86	2018-08	92	88.264264	4.232444	2.641522
2	86	2018-09	100	96.288288	3.854790	2.405824
3	86	2018-10	105	105.315315	0.299401	0.186860
4	86	2018-11	95	96.288288	1.337949	0.835031
5	86	2018-12	93	91.273273	1.891821	1.180709
6	86	2019-01	89	92.276276	3.550508	2.215917
7	86	2019-02	105	92.276276	13.788727	8.605720
8	86	2019-03	108	91.273273	18.325985	11.437481
9	86	2019-04	99	93.279279	6.132895	3.827618
10	86	2019-05	99	101.303303	2.273670	1.419027
11	86	2019-06	92	87.261261	5.430518	3.389256,

`np.float64(1.6022745840356893))`

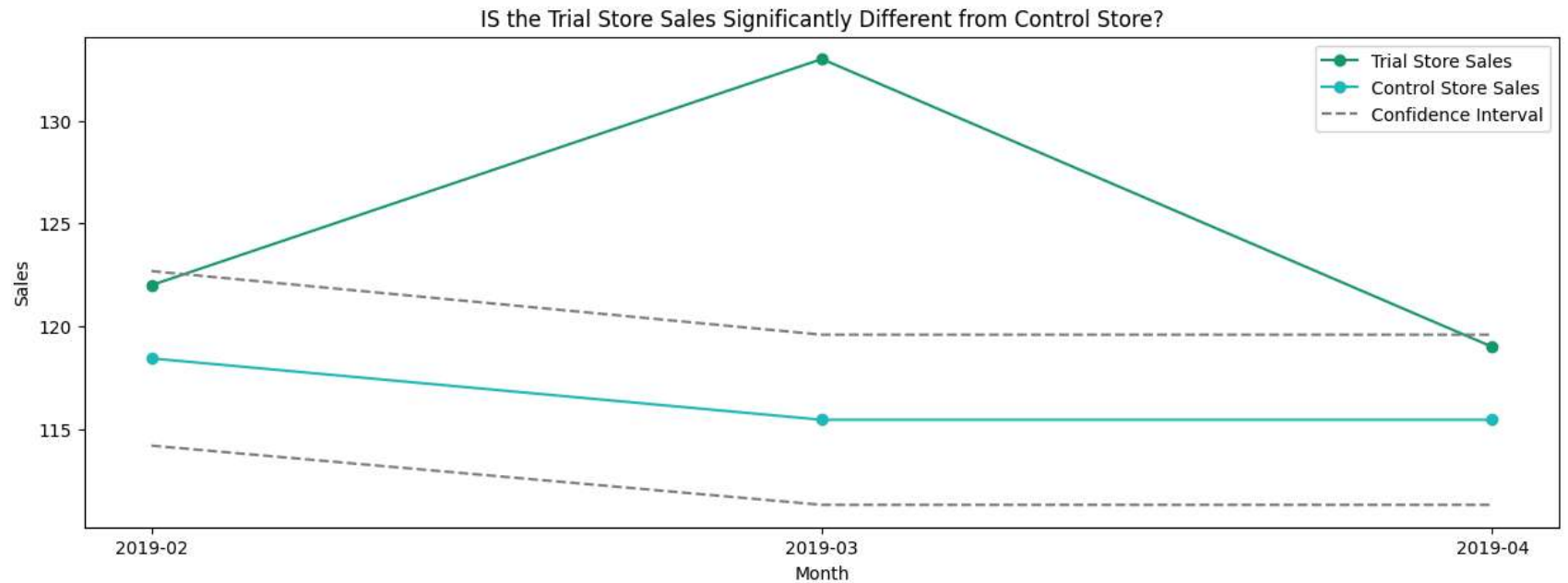


In [699... insight(88,237,'NUM_CUS')

Out[699... (

	STORE_NBR	Month_id	NUM_CUS	controlsales	perc_diff	Ttest
0	88	2018-07	124	124.413146	0.332075	0.185358
1	88	2018-08	125	131.380282	4.856346	2.710713
2	88	2018-09	121	119.436620	1.308962	0.730636
3	88	2018-10	120	117.446009	2.174608	1.213822
4	88	2018-11	123	124.413146	1.135849	0.634008
5	88	2018-12	120	120.431925	0.358646	0.200189
6	88	2019-01	115	110.478873	4.092300	2.284238
7	88	2019-02	122	118.441315	3.004598	1.677105
8	88	2019-03	133	115.455399	15.195999	8.482095
9	88	2019-04	119	115.455399	3.070104	1.713669
10	88	2019-05	123	121.427230	1.295237	0.722975
11	88	2019-06	113	117.446009	3.785577	2.113032,

np.float64(1.791538359841063))



In [700...

```
# Based on these plots
```

```
# we can conclude that trial store has more sales and customers in Feb month ,top in march and decreased in april mor
```

```
# from the plots,the store 77 and 88 gives positive results whereas in store 86 it doesn't seem better growth
```

```
# Based on analysis, i recommend to implement the trial store to increase the sales and customers.
```